Math 356 – Fall 2023
Assignment 5

Question 1

(a)

Let G be a 3-regular graph. Show that if G contains a Hamiltonian cycle then $\chi'(G) = 3$. Let C be a hamiltonian cycle in G, that means C is a path with the same vertex endpoints and $V(C) = V(G)$. Moreover, C is even, since, by the handshaking lemma

$$\sum_{v \in V(G)} \deg(v) = 3|V(G)| = 2 \cdot E(G) \implies |V(G)| \text{ is even}$$

Then there exists a coloring with $E(C) \to \{1, 2\}$, and since no edges of $G - C$ can be incident to the same vertex, by 3-regularity of G, then we can color all edges in $G - C$ with color 3, which yields a 3-edge coloring of G.

(b)

From the lectures on Hamiltonian cycles, we have the canonical counter example of a graph with no Hamiltonian cycle from lecture 7.

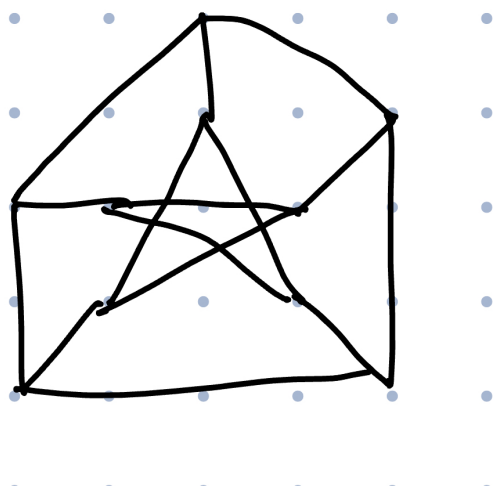

Figure 1: Counter example

By observation, the graph is 3-regular.

(b)

We show first that graphs with no K_4 minor are k-degenerate. Let H be a subgraph of G, then if $H = G$, by lemma 13.2, we have that $\exists v \in V(G)$ such that $\deg(v) \leq 2$. If $H \neq G$, then select $X_1 \subseteq G - H$ to be a clique,

where $|X_1| \leq 2$ in $G - H$, then there exists $v \in G - X$ such that $\deg(v) \leq 2$. If $v \in H$ then we are done, otherwise, consider $G - X_1$, which doesn't have a K_4 minor, since G doesn't have a K_4 minor. Select X_i such that $X_i \subseteq G - X_1 - X_2 \ldots - X_{i-1} - H$, where $G - X_1 - X_2 - \ldots - X_{i-1}$ doesn't have a K_4 minor, until we get a $v \in H$ such that $\deg(v) \leq 2$.